

LSTM-1DResNet: An Intrusion Detection Model for Connected and Autonomous Vehicles Based on Deep Learning

Qiyi He, Yifan Zhang, Ao Xu, Zhiwei Ye, Wen Zhou, Qiao Lin, Tingting Zhang

Abstract—Connected and Autonomous Vehicles (CAVs) are pivotal for enabling intelligent mobility and autonomous driving, yet their inherent connectivity exposes critical vulnerabilities to cyber-attacks, which can precipitate life-threatening accidents, privacy breaches, and systemic failures in transportation networks. Given the escalating deployment of CAVs, ensuring robust cybersecurity is not merely a technical challenge but an urgent safety imperative. Traditional machine learning algorithms, such as Decision Trees (DT) and Support Vector Machines (SVM), often suffer from inadequate feature extraction in high-dimensional network traffic data, significantly compromising the accuracy of cyber-attack identification. Deep learning has emerged as a mainstream solution for Intrusion Detection Systems (IDS) due to its proficiency in handling complex data and automating feature extraction. To address these challenges, we propose LSTM-1DResNet, a novel deep learning-based intrusion detection model comprising an autoencoder and a classifier. The autoencoder innovatively integrates a Long Short-Term Memory Network (LSTM) with a one-dimensional convolutional residual module (Conv1D ResNet), substantially enhancing spatiotemporal feature extraction capabilities for high-dimensional traffic data. The classifier employs a Multilayer Perceptron (MLP) to deliver precise attack classification. The model was evaluated on both the NSL-KDD and CICIDS2017 datasets. On NSL-KDD, LSTM-1DResNet achieved 94.38% accuracy, outperforming standalone CNN and LSTM models by 11%. On CICIDS2017, it achieved 97.98% accuracy, precision of 98.11%, and recall of 97.98%, alongside high F1-scores(96.90%). This dual-dataset validation demonstrates the model's strong potential for enhancing intrusion detection in CAV-related contexts, particularly in addressing high-dimensional feature extraction challenges.

Index Terms—Connected and Autonomous Vehicles, Deep learning, Long Short-Term Memory, Residual Network, Intrusion detection systems.

I. INTRODUCTION

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CAVS are deeply integrated with the external environment through in-vehicle networks, enabling vehicles to share information about their attributes and dynamic information, such as location, speed and driving status, with the Internet. In this way, a high level of information interaction and sharing can be achieved between vehicle and vehicle, vehicle and road, vehicle and human, and vehicle and cloud services. CAVs are developing rapidly due to advances in 4G/5G communication technologies, cellular networks and Bluetooth, as well as the emergence of IoT and big data technologies, further advancing smart mobility applications such as autonomous driving, traffic management and personalised information services [1]. With the development of CAVs, the network data received by vehicles grows exponentially, the network behaviour becomes more complex, and the traditional cybersecurity system gradually fails to meet the demand. Cyber-attacks do not only affect the quality of personalised information services.2020 Tencent Cohen Labs published a Lexus car security report [2], which states that an attacker can exploit security issues with the in-vehicle Bluetooth and diagnostic functions to wirelessly control the car to perform a number of unintended physical operations. This suggests that if an attacker gains control of a vehicle, it would also pose a significant threat to the safety of the driver and passengers. Network Intrusion Detection System (IDS) is a widely used protection technology in the field of network security that enhances the security of a system by monitoring network behaviour to identify malicious or unauthorised actions [3]. Identifying abnormal network behaviour through network intrusion detection is the key to ensure network security and reliability and is an important aspect of the future development of CAVs [4].

Network intrusion detection systems can be categorized into two main types: signature-based and anomaly-based. Signature-based network intrusion detection systems (SIDS) can detect network traffic in real time and identify known network attacks efficiently and accurately. However, SIDS are based on predefined signature libraries whose validity is limited by their timeliness, and SIDS have inherent limitations in detecting unknown threats. Therefore, ensuring the continued effectiveness of the system requires regular maintenance and updating of the signature libraries [5]. Anomaly-based network intrusion detection system (AIDS) identifies intrusion attacks by analyzing the characteristics of network behaviors and still has the ability to protect against unknown attacks. Compared to SIDS, AIDS does not rely on a feature library and is more adaptable and flexible [6]. In the design of

AIDS, machine learning (ML) is widely used, such as DT and SVM techniques, to classify normal and abnormal samples by manually extracting features. However, these methods may negatively affect the protection performance of CAVs due to the increased difficulty of feature processing when facing massive and high-dimensional data. Deep learning (DL) is more adept at handling massive high-dimensional data because of its automatic feature extraction capability, and DL has shown excellent performance in natural language processing (NLP) and computer vision (CV), and is gradually being widely used in AIDS design. Deep learning techniques can extract potential data features in network data to identify network attack behaviors, so DL-based intrusion detection models need to have strong feature extraction capabilities. Currently, many studies have used models such as Convolutional Neural Networks (CNN) [7], Autoencoder (AE) [8], Recurrent Neural Networks (RNN) [9], Long Short-Term Memory Networks (LSTM) [10], Generative Adversarial Networks (GAN) [11], and Deep Belief Networks (DBN) [12] to implement CAV intrusion detection. However, existing deep learning-based IDSs continue to face significant challenges in CAV networks. CAV network traffic constitutes high-dimensional structured data wherein individual traffic samples exhibit dynamic sequential field dependencies (e.g., temporal logic in protocol headers) and fine-grained feature coupling (e.g., anomalous payload patterns), causing substantial overlap between benign behavioral mutations (e.g., transient sensor perturbations) and malicious attacks in the feature space. Current single-model approaches fail to effectively decouple long-range sequential dependencies from deep spatial features: convolutional neural networks (CNNs) capture local spatial patterns but neglect long-distance field dependencies (e.g., cross-protocol-layer anomalies); long short-term memory networks (LSTMs) model sequential relationships yet suffer from high-dimensional feature degradation during recursive propagation. Furthermore, CAVs' real-time operational constraints necessitate low-latency inference with uncompromised accuracy, yet existing methods consistently struggle with the trade-off dilemma between deep feature extraction and computational efficiency, thereby failing to simultaneously achieve robustness and adaptability in complex vehicular networks. Consequently, a mechanism that synergistically optimizes sequential modeling and deep feature extraction is required to enhance feature discriminability while maintaining computational efficiency.

To address the above problems, this paper proposes a deep learning-based LSTM-1DResNet intrusion detection model. Given the characteristics of CAVs network traffic, which include high dimensionality, temporal dependency, and fine-grained feature coupling, the model employs a cascaded feature extraction architecture: first, a LSTM module is used to capture long-range temporal dependencies across fields in the data stream, effectively modeling the dynamic logic between protocol layers; The output is then fed into a one-dimensional convolutional residual network (1DResNet), which uses multi-layer convolutional operations to decouple and enhance deep spatial features. The residual connection mechanism helps mitigate the vanishing gradient problem in deep networks while maintaining low computational overhead, meeting the

real-time requirements of in-vehicle systems. Finally, the extracted high-level features are analyzed by a multi-layer perceptron (MLP) classifier to predict the attack category. Experimental results on the NSL-KDD dataset show that the proposed model achieves approximately 11% better detection performance compared to independent CNN and LSTM models. On the CICIDS2017 dataset, which better aligns with CAVs network environments, the model also demonstrates excellent detection performance (accuracy of 97.98%, F1 score of 96.90%), preliminarily validating its adaptability in diverse network environments. The main contributions of this paper are as follows:

(1) This paper proposes a deep learning framework called LSTM-1DResNet, which combines LSTM with a 1DResNet for intrusion detection in CAVs scenarios. The model uses LSTM modules to capture long-term dependencies between fields in network traffic and further extracts high-dimensional spatial features using residual modules built on Conv1D. Compared with traditional convolutional structures, this approach enhances feature expression capabilities and model training stability to a certain extent.

(2) The feature extraction sequence of this article is to first perform sequence modelling, then deep extraction, placing the Conv1D residual module after the LSTM. This design aims to preserve the integrity of the original data's temporal structure and avoid the potential interference of convolution operations on sequence information, thereby more effectively achieving the staged decoupling and fusion of spatio-temporal features.

(3) The model was evaluated on two public datasets: NSL-KDD and CICIDS2017. The experimental results show that the proposed model achieves an accuracy rate of 94.38% on NSL-KDD and an accuracy rate of 97.98%, a precision rate of 98.11%, a recall rate of 97.98%, and an F1 score of 96.90% on CICIDS2017, verifying its feasibility and effectiveness in processing high-dimensional network traffic data.

The rest of the paper is structured as follows: the second section introduces the related research work; the third section describes in detail the structure of the LSTM-1DResNet model and the roles of each module; the fourth section describes the dataset used in the experiments, the experimental setup, and the experimental results, and analyzes the experimental results; and the fifth section summarizes the work of the paper.

II. RELATE WORK

Many ML methods have been applied in IDS design, such as SVM [13-14], K-Nearest Neighbor (KNN) [15-16], and Random Forest (RF) [17]. However, these methods have high complexity of feature extraction for certain massive and high-dimensional data, which may lead to higher false alarm rate of IDS and lower detection rate of cyber-attacks. Kim et al [18] proposed an intrusion detection system using Genetic Algorithm (GA) to optimize SVM. In this study, GA is combined with SVM to improve the overall efficiency of SVM based IDS. This fusion approach helps in determining the "best detection model" for the SVM classifier and selecting the "best parameters" for the SVM. On the other hand, Panda et al [19] used Naive Bayes (NB) algorithm in their study

TABLE I
SUMMARY OF EXISTING INTRUSION DETECTION METHODS AND THEIR LIMITATIONS

Method	Key Limitations	Dataset	Accuracy
SVM [13]	- Limited handling of high-dimensional network traffic - Models struggle to adapt to the rapid evolution of new types of cyber attacks	KDDCUP'99	89.85%
KNN [15]	- Computationally expensive for real-time detection - Sensitive to hyperparameter selection	NSL-KDD	98.87%
Naive Bayes [19]	- The assumption of feature independence of Naive Bayes deviates from the complex correlation of actual network traffic - High-dimensional data processing is inefficient and cannot meet the performance requirements of real-time intrusion detection	KDDCUP'99 (10%)	96.00%
LSTM [36]	- The false alarm rate is relatively high, which may generate a large number of false alarms in actual deployment and affect system availability	CSE-CIC-IDS2018	99.97%
CNN [37]	- Limited temporal feature extraction - SMOTE introduces synthetic artifacts	AWID	94.00%
CNN-LSTM [38]	- Distributed processing relying on the Apache Spark framework may face challenges in terms of computing resources and real-time performance on resource-constrained in-vehicle devices	NSL-KDD	99.70%
LSTM-CNN [44]	- The model is highly complex and requires significant computing resources, which may pose challenges in balancing performance and efficiency during actual deployment	NSL-KDD	95.70%
		UNSW-NB15	94.90%
		CICIDS-2018	96.70%

to address anomaly detection challenges and evaluated it on KDD Cup dataset. The results show that the NB algorithm exhibits superior performance in terms of maintaining a lower false alarm rate and operational time cost compared to many existing IDS schemes.

Based on the existing research, further exploration also involves the development and application of multiple parallel and hybrid categorization strategies. Yang et al [20] proposed a novel integrated IDS framework called Leader Category and Confidence Decision Integration (LCCDE). The framework is constructed by selecting the best performing model from three state-of-the-art machine learning algorithms (XGBoost, LightGBM, and CatBoost) for each category or attack type. These categorical leadership models and their predictive confidence values are then utilized to accurately identify various types of cyber attacks. Experiments show that the proposed LCCDE framework can be effectively applied to intrusion detection in both in-vehicle and external networks.

Traditional machine learning algorithms have demonstrated a certain degree of effectiveness in identifying cybersecurity threats, with detection accuracy that meets basic requirements under specific conditions. However, as CAVs become increasingly widespread, the dynamic and complex nature of network environments has significantly intensified. The limitations of traditional machine learning methods in feature processing capabilities have become increasingly apparent, particularly when dealing with high-dimensional, unstructured

data. Their detection accuracy is easily affected by changes in the environment, and controlling false positive rates poses significant challenges. Therefore, to adapt to the complex network environment of CAVs, it is necessary to further optimise the performance of IDS to improve detection accuracy and reduce false positive rates. This research direction has been preliminarily argued in relevant literature [21-22]. In this context, enhancing the feature extraction capabilities of models is considered one of the key paths to improving the protective effectiveness of IDS.

DL has provided feasible solutions for NLP [23-25], CV [26-27], and speech recognition [28-30] and other fields, thanks to the core neural network (NN) technology of DL, which has the ability to perform hierarchical representation and feature learning, as well as process high-dimensional data and extract effective information. Therefore, DL has significant advantages when dealing with high-dimensional and time-series network traffic data. Specifically, deep learning models can automatically extract multi-level features from high-dimensional raw network traffic, effectively reducing reliance on manual feature engineering, and capturing complex non-linear attack patterns. Additionally, it supports joint modelling of spatio-temporal features, enabling the simultaneous capture of local spatial anomalies within packet payloads and long-range temporal dependencies across packets, thereby providing a more comprehensive understanding of attack behaviour. This capability endows the model with robust nonlinear fitting and

generalisation performance, enabling it not only to accurately identify known attacks but also to discover unknown or variant attacks with limited training data, significantly enhancing the adaptability and robustness of intrusion detection systems in addressing new network threats [41]. Early studies attempted to combine shallow learning and deep learning techniques [31]. For example, [32] proposed an information entropy-based deep belief network model (IE-DBN), which uses information gain (IG) to reduce the dimensionality of high-dimensional data features to eliminate redundant features, and determines the number of hidden layer neurons and network depth of the DBN network based on information entropy, thereby optimising the model structure. Anzer et al. employed a multi-layer perceptron (MLP) neural network to detect abnormal behaviour in internet environments. Their research results were presented in the form of predicted outputs, classification reports, and confusion matrices. Simulation analyses indicated that this method has a certain degree of effectiveness under specific conditions [33].

In addition, deep learning models such as LSTM [34] and CNN [35] have gradually been applied in intrusion detection research. LSTM is used to mitigate the vanishing gradient and exploding gradient phenomena that traditional recurrent neural networks (RNNs) may encounter when processing long-term dependencies, thereby improving their ability to learn from long sequence data to a certain extent. CNNs demonstrate certain applicability in local feature capture and pattern recognition. For example, [36] adopted a network intrusion detection system (WNIDS) model based on a bidirectional long short-term memory network (BiLSTM), which demonstrated certain detection capabilities for network intrusion behaviour under specific experimental conditions. Research by [37] designed a network architecture that integrates CNN into the data processing layer and combines synthetic minority class oversampling technique (SMOTE) to optimise the data processing workflow, aiming to enhance the stability of the learning process. During the testing phase, a nearest neighbour classifier is used to identify new attack types. Experimental results based on the AWID dataset indicate that the model demonstrates certain detection accuracy advantages over SVM and RF methods under specific scenarios. To build a more adaptive IDS, combining LSTM and CNN methods is one of the directions that researchers are exploring. For example, Alferaidi et al. [38] proposed an intrusion detection model for vehicle networks (AA distributed combined deep learning model) based on the Apache Spark distributed computing framework. This model integrates CNN and LSTM to extract and analyse features from large-scale data, aiding in the identification of intrusion behaviour and monitoring of abnormal activities. Sun et al. [40] proposed a CNN-LSTM model with attention mechanism (CLAM), which uses one-dimensional convolution (Conv1D) to extract signal features at time nodes and inputs them into a bidirectional LSTM to capture temporal dependencies; the attention mechanism calculates the weight distribution of the bidirectional LSTM hidden states and performs weighted focusing on key time steps. Experimental analysis indicates that this method may help accelerate model convergence and improve prediction

accuracy under specific conditions. Xue et al. [44] proposed the HAE-HRL model, which combines a hybrid autoencoder (HAE) for feature selection and designs a residual network-based LSTM-CNN hybrid classifier (HRL), achieving high-performance detection on multiple datasets such as NSL-KDD, UNSW-NB15, and CIC-IDS-2018 datasets, highlighting the synergistic value of feature optimisation and deep hybrid architectures.

Existing research has demonstrated that the combination of LSTM and CNN can effectively identify network attack behaviour (Table I). This hybrid architecture combines the spatial feature extraction capabilities of CNN with the temporal modelling capabilities of LSTM, providing intrusion detection systems with a more comprehensive feature representation. The proposed LSTM-1DResNet model adopts a 'sequence-first, spatial-second' feature extraction order: first, the LSTM module processes the input data to fully capture the long-range temporal dependencies across fields in network traffic; subsequently, 1DResNet is used for deep feature decoupling, where Conv1D replaces traditional Conv2D to more efficiently process one-dimensional network traffic data. This design choice is based on an analysis of the characteristics of CAVs network traffic. Placing the LSTM at the front end ensures the integrity of the temporal structure of the raw data and avoids potential interference with sequence information caused by convolution operations in the early stages, thereby more effectively achieving the staged decoupling and fusion of spatio-temporal features [45]. Additionally, the introduction of residual modules not only enhances feature extraction capabilities but also improves model training stability through a skip connection mechanism.

III. MODEL

The LSTM-1DResNet intrusion detection model proposed in this paper, aimed at detecting cyber-attacks on CAVs, comprises autoencoder and classifiers, with its architecture depicted in Fig.1. The principal objective of securing the network within the CAV environment is the identification of cyber-attack behaviors; consequently, the LSTM-1DResNet model is configured as a binary classification model. This section introduces the LSTM and residual module components utilized by LSTM-1DResNet and provides a detailed exposition of the LSTM-1DResNet model.

A. Long-term Memory Network

LSTM is a modified RNN method designed to alleviate the gradient explosion and vanishing problems. Traditional RNNs suffer from the vanishing gradient problem, which hinders the modelling of long-term dependencies. LSTMs introduce gated memory units to retain key temporal features. Specifically, the forget gate regulates the retention of information from previous states, the input gate controls the integration of new features, and the output gate determines state propagation, thereby jointly achieving the persistent storage of attack-related patterns across extended sequences. The structure of the LSTM cell is shown in Fig.4.

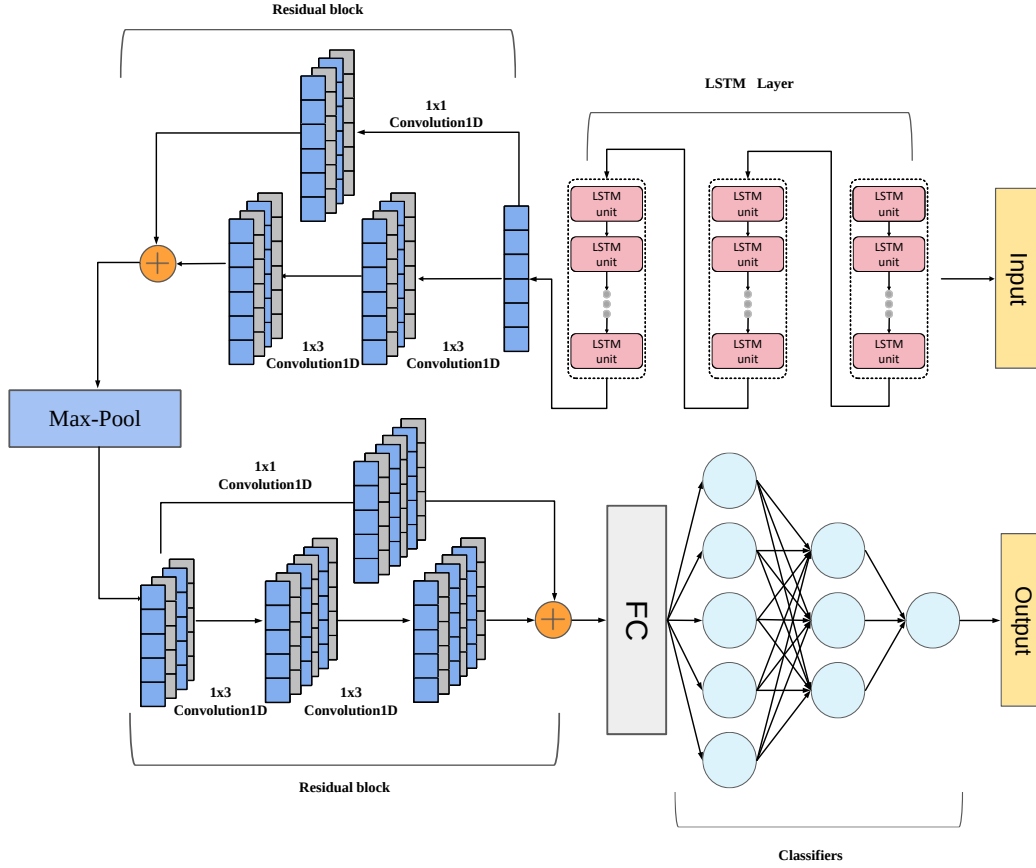


Fig. 1. Structure of LSTM-1DResNet.

The key function of LSTM is to record long-term information. The forgetting gate controls the selective forgetting of information in the cell state, the input gate selectively adds new information to the cell state, and the output gate selectively outputs information. At time point t , the LSTM cell processes the cell state $C_t - 1$ at the previous time point $t - 1$ through the forgetting gate, input gate and output gate, and then generates the cell state C_t at the current time point t and passes C_t to the next cell.

LSTM excels in extracting long-distance dependency features. Through its gating mechanism, LSTM is able to effectively capture and maintain long term dependencies in sequential data and thus extract key features, which is crucial for accurately identifying complex network attack patterns. By retaining key features in sequence data, LSTM is able to effectively distinguish data with attack behaviours, thus improving the detection performance of intrusion detection systems.

B. 1DResNet

In deep learning, increasing the number of hidden layers is a simple and effective method for improving model performance. However, with the increase in the number of hidden layers, the problem of gradient vanishing or gradient explosion is prone to occur, and even model degradation

can occur. The residual network (ResNet) [46] proposed by He Kaiming's team in 2015 innovatively introduces shortcut connections to skip one or more layers, transforming the $f(x) = x' - x_1$ problem into the $x' = f(x) + x_1$ problem, which effectively alleviates these problems. Fig.2(a) shows the residual structure more visually. Inspired by ResNet, in this paper, the convolutional layer using Conv2D is replaced by the convolutional layer using Conv1D, and a 1x1 convolutional layer is added to the shortcut connection (shown in Fig.2(b)), where the 1x1 convolution serves to adjust the number of channels of the input to match the output of the convolutional layer.

The advantage of using 1D convolution(Fig.3(b)) instead of 2D convolution(Fig.3(a)) is that 1D convolution is more suitable for processing one-dimensional sequence data and can extract local features in sequence data more efficiently. Compared with 2D convolution, 1D convolution is also more efficient in terms of computational complexity and resource consumption. Especially when dealing with large-scale 1D data, it can improve the running speed and processing capability of the model.

In the LSTM-1DResNet proposed in this paper, it uses LSTM to extract long-range dependent features, uses the residual module to further extract data features, and it mitigates the risk of overfitting and improves the depth and stability

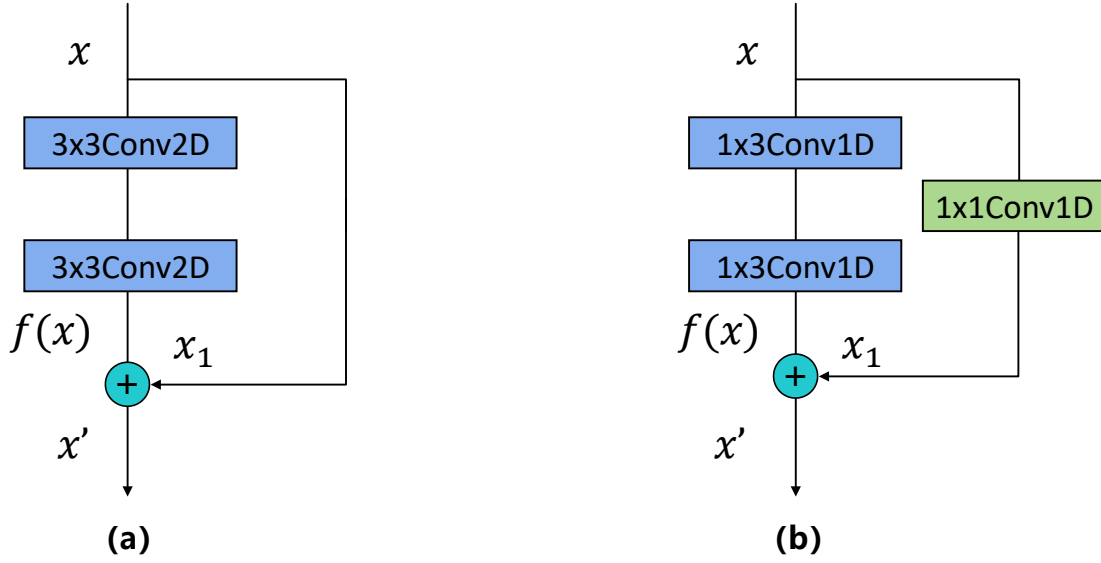


Fig. 2. (a)Residual block in ResNet, (b)Residual block in LSTM-1DResNet.

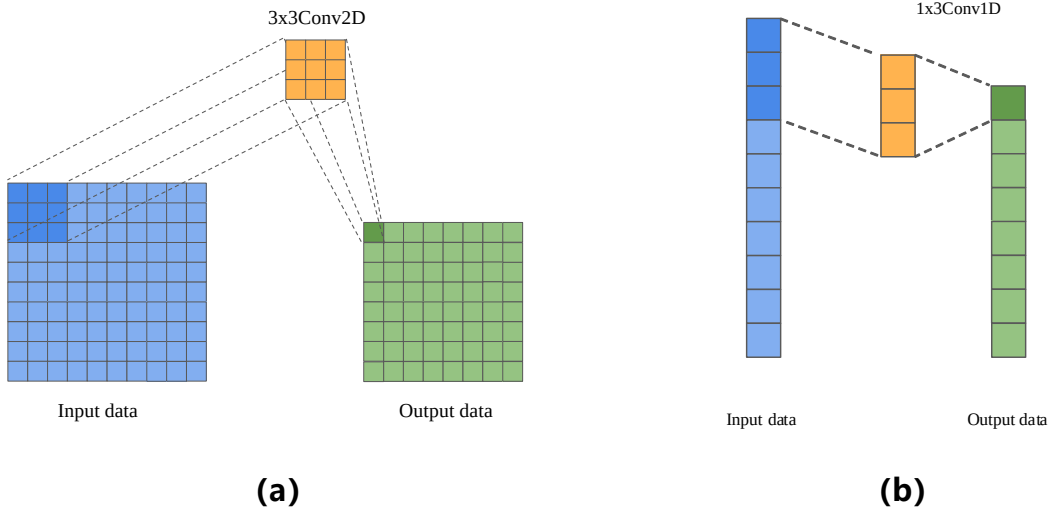


Fig. 3. Structure of LSTM-1DResNet.

of the model. The residual module alleviates the degradation problem of deep networks through identity shortcuts, reconstructing layer learning as residual functions. This mechanism maintains representation capabilities while preserving gradient flow during backpropagation, which is particularly critical for hierarchical extraction of multi-scale attack features in intrusion detection.

C. LSTM-1DResNet

As shown in Fig.1, the first component of the model is composed of LSTM units. In order to adapt to the input format of LSTM, we add a dimension to the input data x and adjust the order of the dimensions to $(batchsize, length, channels)$,

where $length$ is the length of a single piece of data, and $channels$ is the number of channels in the data, which is set to 1 by default. After this processing, the data flows to the LSTM layer, capturing the sequential dependencies between different feature dimensions in the data. After processing by the LSTM layer, the output of the last time step of each sequence is selected to form a new output data dimension $(batchsize, length)$.

Then, the data output from the LSTM is reconstructed into $(batchsize, channels, length)$ after which it is fed into the residual layer to further extract data features. The residual layer consists of two residual modules, each with the same structure (Fig.2(b)). The backbone of the residual module

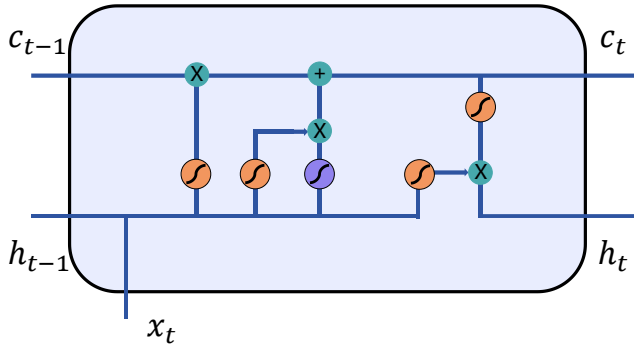


Fig. 4. Structure of LSTM unit.

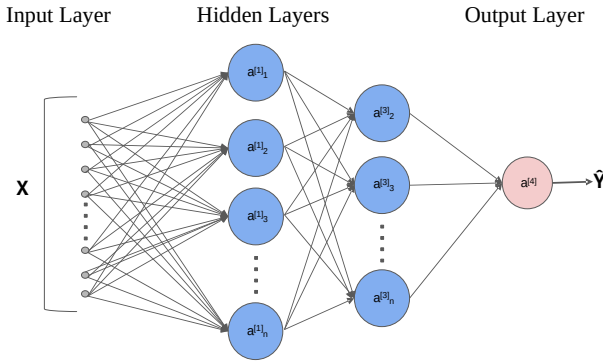


Fig. 5. Structure of MLP.

contains two convolutional layers using 1×3 convolutional kernels, where the first layer is responsible for extracting the base features from the raw data, and the second layer performs feature fusion and abstraction on top of the base features to further extract more advanced features. The fast connectivity boosts the channel dimension of the data through a convolutional layer using a 1×1 convolutional kernel with the aim of matching the output of the backbone for residual operations. In the first residual module of the residual layer, the channel dimension of the data will be increased from 1 to 32, and then the data will be downsampled by a one-dimensional maximum pooling layer. Downsampling serves to enhance local features and reduce computational complexity. In the second residual module, the feature dimension of the data will be further extended to 64 and feature extraction is completed after the data passes through the residual layer. The output of the residual layer is converted into a one dimensional vector using Flatten layer which is used as an input to the classifier. The classifier is an MLP consisting of fully connected layers(Fig.5).

As shown in Fig.5, the classifier has an inverted pyramid structure. The inverted pyramid structure helps the network learn the mapping from raw inputs to higher-order abstract features by reducing the number of nodes layer by layer, which can effectively reduce the risk of overfitting and reduce the network complexity. It makes the model more lightweight and efficient while maintaining good generalisation perfor-

mance. In the classifier, the vectors are first converted to 32 dimensions and then further compressed to 16 dimensions, which is a series of linear operations aimed at accurately filtering and compressing the feature information that is crucial to the classification task. Finally, the predictions are output through the output layer and processed by a Sigmoid activation function to ensure that all output values are within the range (0,1).

IV. EXPERIMENTAL

TABLE II
COMPOSITION OF THE NSL-KDD DATASET

Class	Train	Percentage	Test	Percentage
Normal	67343	53.458%	9710	43.075%
DoS	45927	36.458%	7457	33.080%
Probe	11656	9.253%	2421	10.740%
R2L	995	0.790%	2754	12.217%
U2R	52	0.041%	200	0.887%
Total	125973	100%	22545	100%

A. NSL-KDD dataset

The NSL-KDD dataset is obtained based on the KDD-cup99 dataset, which is further optimised for the KDDcup99 dataset. KDDcup99 contains a large number of redundant and duplicate records, and the NSL-KDD dataset is very useful in removing redundant and duplicate records from the KDDcup99 dataset. Therefore, most of the intrusion detection researches have used NSL-KDD dataset as the training and testing dataset. sets as training and test datasets. NSL-KDD provides training dataset KDDTrain and test dataset KDDTest with the number of 125,973 and 22,544, respectively. Each data point has 41 attributes (3 nominal attributes, 6 binary attributes, and 32 numeric attributes), which represent different characteristics of the network flow. As well as labels indicating the type of attack or normal behaviour. For the attack types, there are four different attack modes:

- 1) Denial of Service Attack (DoS): DoS is an attack that consumes resources by sending a large amount of traffic to a target system so that it is unable to process legitimate network traffic or service access;
- 2) Probing: In a probing attack, the attacker aims to obtain information about the target system (e.g., scanning for ports in use and scanning for IP addresses);
- 3) Remote to Local (R2L): R2L is an attack that attempts to gain local access to a remote machine by sending spoofed remote traffic to the target. Actions such as password guessing and HTTP tunneling are considered R2L attacks;
- 4) Use of Root Users (U2R): In the case of U2R, the attacker first gains access to the target system as an honest user and then gains root privileges by causing system failures (e.g., buffer overflows and rootkits).

TABLE III
COMPOSITION OF THE CICIDS2017 DATASET

File Name	Type of Traffic	Number of Records
Monday-WorkingHours.pcap_ISCX.csv	Benign	529,918
Tuesday-WorkingHours.pcap_ISCX.csv	Benign	432,074
Wednesday-WorkingHours.pcap_ISCX.csv	SSH-Patator	5,897
	FTP-Patator	7,938
	Benign	440,031
	DoS Hulk	231,073
	DoS GoldenEye	10,293
	DoS Slowloris	5,796
	DoS Slowhttptest	5,499
	Heartbleed	11
Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv	Benign	168,186
	Web Attack-Brute Force	1,507
	Web Attack-Sql Injection	21
	Web Attack-XSS	652
Thursday-WorkingHours-Afternoon-Infiltration.pcap_ISCX.csv	Benign	288,566
	Infiltration	36
Friday-WorkingHours-Morning.pcap_ISCX.csv	Benign	189,067
	Bot	1,966
Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv	Benign	127,537
	Portscan	158,930
Friday-WorkingHours-Afternoon-DDoS.pcap_ISCX.csv	Benign	97,718
	DDoS	128,027
Total Instance/Record		2,830,743

Table II illustrates the entire distribution of the NSL-KDD dataset over normal and attack categories. As can be seen from Table 1, even though the NSL-KDD dataset is obtained by improving the KDDcup99 dataset, there is still a large gap in the amount of data in different attack categories. However, when the classification conditions are set to normal and attack types for binary classification, 53.46% of the NSL-KDD dataset is normal type data and 46.54% is attack type data, and its data distribution maintains a certain balance, so in this paper, the NSL-KDD dataset is used to train and test the binary classification performance of the model.

B. CICIDS2017 dataset

The CICIDS2017 dataset, released by the ISCX/CIC consortium, is a widely recognised benchmark encompassing contemporary network traffic and attack scenarios. The MachineLearningCSV split is adopted, comprising eight traffic-monitoring sessions stored as comma-separated value (CSV) files. The corpus contains benign traffic and 14 distinct attack categories, as summarised in Table III. CICIDS2017 provides rich flow-level attributes suitable for characterising modern and sophisticated behaviours. For example, Subflow Fwd Bytes and Total Length of Fwd Packets are critical for detecting Infiltration and Bot attacks; Bwd Packet Length Std is informative for DDoS, DoS Hulk, DoS GoldenEye, and Heartbleed; Init Win Fwd Bytes is relevant to Web-Attack, SSH-Patator, and FTP-Patator; while Min Bwd Packet Length and Fwd Average Packet Length assist in recognising benign traffic. Owing to its fidelity to real-world traffic and threat

characteristics, CICIDS2017 is widely adopted for evaluating intrusion-detection methodologies.

C. Dataset preprocessing

In order for the model to better extract data features and enhance the model prediction performance, Dataset was pre-processed before feeding the data into the model:

1) *Feature Processing*: Convert the feature values of character types to numerical types through unique thermal encoding, and unify the feature scales through standardization methods to ensure that all features are in the same magnitude, which is beneficial for model understanding and training.

2) *Feature selection*: A feature selection method using recursive elimination removes redundant features and selects a subset of features that are more relevant to the classification result for training and prediction.

These steps help to improve the feature extraction ability of the model on the data and enhance the classification performance of the model. In addition, the training set is divided into a training set and a validation set in the ratio of 7:3 to test the performance of the model during the training process.

D. Evaluation indicators

In this paper, the commonly used Accuracy, Precision, Recall and F1 scores are used as evaluation metrics to evaluate the performance of all models.

1) *Accuracy*: Accuracy measures the proportion of total samples that are correctly predicted by the model. Accuracy is an intuitive and commonly used metric for binary classification problems.

2) *Precision*: Precision measures the proportion of positive cases predicted by the model that are actually positive. For intrusion detection systems, Precision reflects the accuracy of the system's prediction of intrusion behavior. Higher Precision means fewer false alarms (misreporting normal behavior as intrusion), which reduces unnecessary alarm processing.

3) *Recall*: Recall measures the proportion of actual positive case samples that are correctly predicted as positive cases by the model. For intrusion detection systems, Recall reflects the system's ability to recognize actual intrusions. A higher Recall means fewer missed alarms (misreporting intrusions as normal behavior), which implies a higher level of security for the system.

4) *F1*: The F1 score is the reconciled average of Precision and Recall, which takes into account the balance between the two and can reflect the overall performance of the model more comprehensively. For intrusion detection systems, the F1 score can reduce false alarms while ensuring as much as possible that the system can correctly identify the actual intrusion behavior.

In the experiments in this paper, normal data is labeled as 0 and abnormal data is labeled as 1. The LSTM-1DResNet model outputs either 0 or 1, where 1 indicates that the data is predicted to be abnormal and 0 indicates that the data is predicted to be normal. Abnormal data is true positive (TP) if it is correctly predicted to be 1. If abnormal data is incorrectly predicted to be 0, it is called false negative (FN); normal data is correctly predicted to be 0 is true negative (TN); and if normal data is incorrectly predicted to be 1, it is called false positive (FP).

In the experiment, the confusion matrix is constructed based on the prediction results of the model on the test set, and the four evaluation metrics of Accuracy, Precision, Recall and F1 score are calculated. The evaluation metrics are calculated as follows:

$$\text{Accuracy} = \frac{TP + TN}{\text{total}} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1} = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} \quad (4)$$

E. Experimental set

The experiments were conducted using pytorch to build the model, running in Ubuntu system, with hardware using 16G RAM and NVIDIA GTX3060 GPU. the activation function used the relu activation function, the learning rate was set to 0.0001, and the model was trained for 100 epochs at a time.

F. Experimental results and analysis

The evaluation conducted on NSL-KDD and CICIDS2017 datasets provides critical insights for intrusion detection in Connected and Autonomous Vehicle environments (Table IV).

Traditional machine learning approaches demonstrate significant limitations in handling high-dimensional network traffic, as evidenced by SVM's performance of 75.38% accuracy and 70.62% F1-score on the NSL-KDD dataset. These results confirm the inadequacy of conventional methods for extracting complex patterns inherent in vehicular network communications. Deep learning models offer moderate improvements, with one-dimensional convolutional neural networks achieving 81.41% accuracy and long short-term memory networks reaching 82.48% accuracy on NSL-KDD. However, both approaches exhibit constrained performance in comprehensive spatiotemporal feature extraction, particularly when processing the multi-layer protocol dynamics characteristic of CAV traffic.

Contemporary hybrid architectures present notable trade-offs that warrant careful consideration. The CNN-GRU model achieves 93.10% accuracy on NSL-KDD but shows reduced effectiveness on CICIDS2017 with 90.17% accuracy, indicating potential dataset-specific limitations. Similarly, JAYA-LSTM demonstrates high recall of 97.68% on NSL-KDD yet suffers from precision of only 88.11%, suggesting challenges in minimizing false positive detections—a critical concern for safety-critical vehicular systems. The LSTM-CNN architecture reports the highest NSL-KDD accuracy at 95.70%, but this performance comes at substantial computational cost. Its complex design incorporating hybrid autoencoders, six convolutional layers, and self-attention mechanisms requires 447.10 million floating-point operations per second and 80,000 parameters, which may create significant barriers for deployment in resource-constrained vehicular environments.

In contrast, the proposed LSTM-1DResNet model achieves 94.38% accuracy on NSL-KDD, representing improvements of 12.97 percentage points over one-dimensional convolutional neural networks and 11.90 percentage points over standalone long short-term memory networks. On the CICIDS2017 dataset, the model demonstrates robust performance with 97.98% accuracy, 98.11% precision, 97.97% recall, and 98.00% F1-score. Crucially, this performance is attained with only 27.8 million floating-point operations per second and 220 parameters, establishing a 16.1-fold reduction in computational complexity compared to the LSTM-CNN architecture. This efficiency profile positions the model as a viable candidate for embedded deployment in vehicular systems, where balancing detection accuracy with real-time processing constraints remains a persistent challenge. The consistent high performance across both datasets suggests the architecture's adaptability to diverse network environments.

G. Ablation Experiment

The ablation study systematically evaluates two core architectural components through comparative analysis presented in Table V and visualized in Fig. 6 and Fig. 7. First, the sequence-first configuration, where long short-term memory processes temporal dependencies before one-dimensional residual network feature extraction, demonstrates measurable advantages over the spatial-first alternative. On the NSL-KDD dataset, the sequence-first approach achieves 94.38% accuracy with 6956 correctly classified normal traffic instances and 1652 correctly

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Method	NSL-KDD				CICIDS2017			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
SVM	0.7538	0.7548	0.7598	0.7062	0.9312	0.8996	0.8793	0.8890
1DCNN	0.8141	0.8377	0.8300	0.7680	0.9692	0.9358	0.9744	0.9535
LSTM	0.8248	0.8432	0.8370	0.7782	0.9587	0.9167	0.9658	0.9384
CNN-GRU [42]	0.9310	0.9375	0.9029	0.9234	0.9017	0.9234	0.9124	0.9205
JAYA-LSTM [43]	0.9200	0.8811	0.9768	0.9411	0.9421	0.9811	0.9251	0.9111
CNN-LSTM [39]	0.9225	0.8930	0.9804	0.9349	-	-	-	-
LSTM-CNN [44]	0.9570	0.9660	0.9580	0.9570	-	-	-	-
LSTM-1DResNet	0.9438	0.9262	0.9047	0.8675	0.9798	0.9811	0.9797	0.9800

TABLE V
PERFORMANCE COMPARISON OF ABLATION EXPERIMENT

Method	NSL-KDD				CICIDS2017			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
LSTM-1DResNet	0.9438	0.9262	0.9047	0.8675	0.9798	0.9811	0.9797	0.9800
1DResNet-LSTM	0.92938	0.91770	0.86711	0.84199	0.9788	0.9519	0.9866	0.9677
LSTM-1DCNN	0.92116	0.91167	0.84728	0.82712	0.9759	0.9486	0.9800	0.9632

identified attack instances, compared to 92.94% accuracy for the spatial-first approach with 6979 correctly classified normal instances but only 1497 correctly identified attacks. The confusion matrices reveal that the sequence-first configuration reduces false negatives by 55 instances (189 vs. 166 false negatives for normal traffic) while maintaining comparable false positive rates (323 vs. 323 for attack traffic), resulting in a more balanced precision-recall profile. This pattern persists in the CICIDS2017 evaluation, where the sequence-first model correctly identifies 165,272 attack instances with only 15,443 false negatives, compared to 166,631 correctly identified attacks but 17,620 false negatives for the spatial-first approach.

The loss curves provide critical insights into the training dynamics of these configurations. On the NSL-KDD dataset, the spatial-first model initially converges quickly with the training loss but gradually diverges from the optimal trajectory after approximately 30 epochs, indicating reduced stability in later training stages. In contrast, the sequence-first model maintains consistent alignment between training and validation losses throughout the entire training process. On the CICIDS2017 dataset, the spatial-first model consistently demonstrates poorer alignment between training and validation losses, failing to achieve the same level of convergence as the sequence-first approach, which maintains stable loss trajectories across all training epochs.

Second, the contribution of the residual module is rigorously assessed by comparing the full LSTM-1DResNet model against an ablated version using standard one-dimensional convolutional layers (LSTM-1DCNN). On NSL-KDD, the residual-enhanced model correctly classifies 1,652 attack instances with only 189 false negatives, while the non-residual variant identifies only 1,418 attacks with 161 false negatives. The confusion matrices further reveal that the residual model maintains superior precision by reducing false positives from 558 to 323 while preserving high recall. This advantage becomes even more pronounced in the CICIDS2017 evaluation, where the residual model correctly identifies 165,272 attacks with 15,443 false negatives, compared to 164,762 correct identifications with 18,198 false negatives for the non-residual variant.

The loss curve analysis reveals distinct training patterns between these variants. On the NSL-KDD dataset, the LSTM-1DCNN model demonstrates reasonable convergence during early training stages but exhibits increasing oscillation amplitude in validation loss during the later training phase, indicating reduced stability as training progresses. In contrast, the residual-enhanced model maintains consistent and stable loss trajectories throughout the entire training process. On the CICIDS2017 dataset, the LSTM-1DCNN model shows substantial oscillations in validation loss both during early and

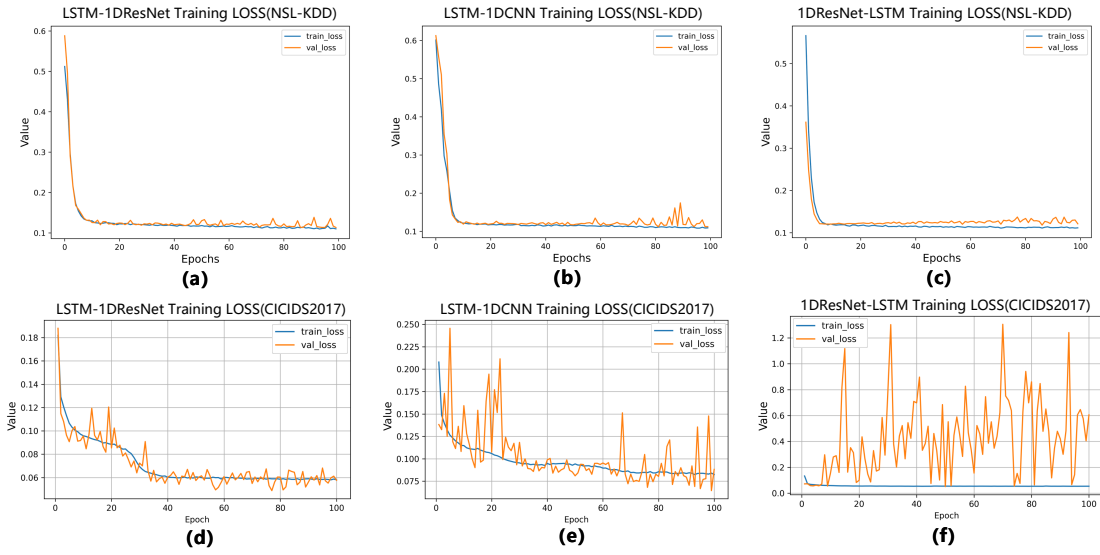


Fig. 6. Loss curve of ablation experiment.

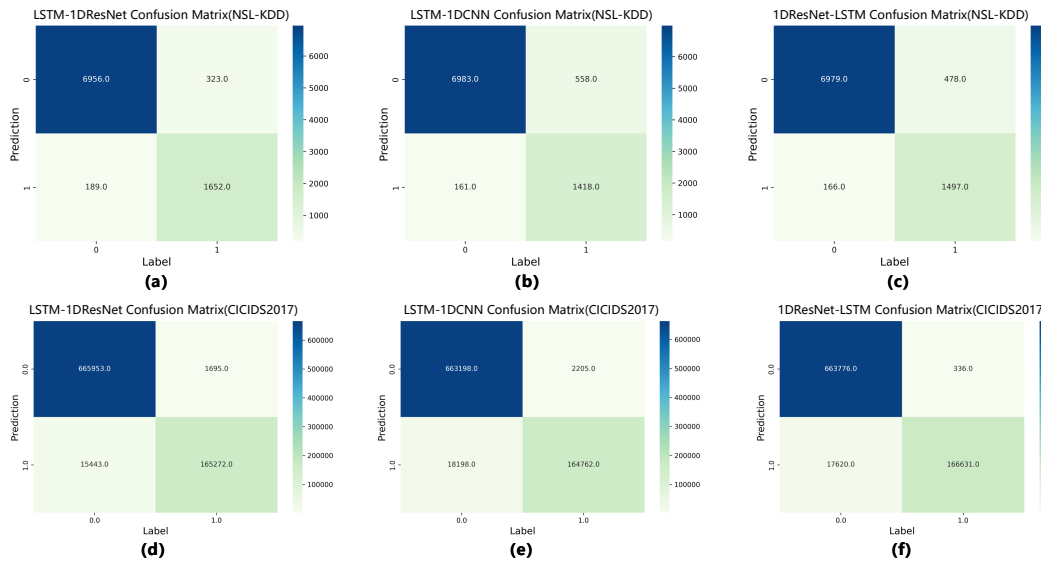


Fig. 7. Confusion matrix of ablation experiment.

late training stages, suggesting fundamental instability in this configuration when processing more complex network traffic patterns. The residual-enhanced model, however, maintains stable convergence across all training epochs, demonstrating superior generalization capability across diverse network environments.

H. Discussion

The results highlight important considerations for CAV intrusion detection. The computational efficiency of LSTM-1DResNet is 27.8 M FLOPS, with an average inference latency of 0.0242 milliseconds (24.23 microseconds) per sample, indicating that LSTM-1DResNet is feasible for practical deployment in resource-constrained vehicular environments

where latency constraints typically require sub-100 millisecond response times. The observed performance differential between sequence-first and spatial-first architectures supports the rationale for prioritizing temporal dependency modeling in CAV traffic analysis, as protocol-layer dynamics exhibit strong sequential characteristics. Notably, the model achieves 97.98% accuracy on CICIDS2017—a dataset featuring modern attack patterns relevant to vehicular networks—with precision and recall exceeding 97.97%, indicating robust capability in distinguishing normal and malicious traffic. While LSTM-CNN [44] reports marginally higher NSL-KDD accuracy (95.70% vs. 94.38%), its substantially higher computational requirements (447.10 M FLOPS) may limit practical deployment in embedded vehicular systems. The consistent performance across both

datasets (94.38% on NSL-KDD; 97.98% on CICIDS2017) demonstrates the framework's adaptability to diverse network environments, though further validation in real-world CAV testbeds remains warranted. The 11.90 percentage point improvement over standalone LSTM on NSL-KDD quantifies the efficacy of the cascaded architecture in addressing high-dimensional feature extraction challenges, while the ablation study confirms the critical role of both the sequence-first design and residual connections in optimizing model behavior. These findings collectively suggest that balanced architectural design can effectively address the tension between detection performance and computational feasibility in vehicular security systems.

V. CONCLUSION

Connected and autonomous vehicles (CAVs) critically enhance intelligent mobility yet introduce severe cybersecurity vulnerabilities that threaten safety-critical operations. While deep learning-based intrusion detection systems (IDS) offer promise, existing models remain constrained by inadequate spatiotemporal feature extraction and prohibitive computational demands for vehicular deployment. This work proposes LSTM-1DResNet, a novel end-to-end architecture integrating LSTM for long-range temporal dependency modeling and a one-dimensional convolutional residual network (1DResNet) for spatial feature refinement. The model's cascaded design—processing sequential dependencies before spatial extraction—preserves temporal integrity while avoiding convolution-induced distortion of protocol-layer dynamics, a critical limitation in prior hybrid architectures. Validated on both NSL-KDD and CICIDS2017 datasets (the latter better reflecting modern CAV network characteristics), LSTM-1DResNet achieves 94.38% accuracy on NSL-KDD and 97.98% accuracy, 98.11% precision, and 97.98% recall on CICIDS2017. Crucially, it accomplishes this with only 27.8 M FLOPS and 220 parameters, enabling real-time operation within vehicular latency constraints (<100 ms). Despite these advances, the severe class imbalance in current datasets (e.g., DoS attacks at 52.9% versus U2R attacks at 0.0009% in NSL-KDD) risks bias toward prevalent attack types. Future work will develop a robust end-to-end multi-class IDS incorporating adaptive focal loss and GAN-based minority-class augmentation to ensure reliable detection of rare cyberattacks while maintaining ultra-low computational overhead for safety-critical CAV environments.

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